

Week 01: MATH 350 Recap

CSU, Chico Math 351

2026-01-23

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MATH 350 Topics to Recall

In MATH 350, a few topics covered included:

- ▶ Set Operations, Conditional Probability, Independence
- ▶ Random Variables: Discrete and Continuous

To sum up: Populations are described with functions that produce random data, letting us estimate characteristics about the population.

These refresher notes go back over random variables to set the stage for our next topic: maximum likelihood estimation.

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Random Variables

Statistics is formed around the idea of a **random variable**: a function that maps outcomes of the sample space to a number.

Numerical variables can be **discrete**, and described with a Probability Mass Function (PMF)...

or **continuous**, and described with a Probability Density Function (PDF).

Discrete Random Variable

We theorize that data are generated from a function that only takes on mass at each value in the sample space. Probability statements are made from the area under the function.

PMF reminders:

- ▶ $P(X = x) \geq 0$, for all x
- ▶ $\sum_{x \in \mathcal{S}_X} P(X = x) = 1$
 - ▶ re CDF, $F_X(\infty) = 1$.
- ▶ $P(a < X \leq b) = \sum_{a < x \leq b} P(X = x) = F(b) - F(a)$
- ▶ watch the inequalities, $P(X \leq 3) \neq P(X < 3) = P(X \leq 2)$

Cumulative Distribution Function (CDF)

Probabilities of events of the form $\{X \leq x\}$ are defined generally by the term **cumulative distribution function** (CDF) for both discrete and continuous random variables.

cumulative distribution function

The CDF of a random variable X is defined to be

$F_X(x) = P(X \leq x)$, for all numbers x .

Probability Density Function (PDF)

The PDF of a continuous random variable X is a nonnegative function f_X under which we measure probability on intervals of interest, say $[a, b]$.

$$P(a < Y < b) = \int_a^b f_Y(y) dy.$$

- ▶ $f_Y(y) \geq 0$, for all y
- ▶ $\int_{-\infty}^{\infty} f_Y(y) dy = 1$
 - ▶ re CDF, $F_Y(\infty) = 1$.
- ▶ $P(Y = y) = 0$, for all y
- ▶ $P(a < Y \leq b) = \int_a^b f_Y(y) dy = F(b) - F(a)$
- ▶ inequalities don't matter, $P(Y \leq 3) = P(Y < 3)$

Cumulative Distribution Function (CDF)

The CDF of the continuous random variable Y is defined similarly:

$$F_Y(y) = \int_{-\infty}^y f_Y(s) ds.$$

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Motivating Parameters

As a note, all the sample statistics you've ever heard of are first defined for populations and then translated into data-terms. The population (not sample) mean is called the **expected value**. We should think about the center of the function that describes the random variable of interest.

For instance, a coin flip's function is

$$f(x|p) = p^x(1 - p)^{1-x}, \text{ for } x \in \{0, 1\}.$$

The measure of center for this function is probably more easily seen with a table or a graph.

x	1 (Heads)	0 (Tails)
$P(X = x)$	p	$1 - p$

Expected Value, discrete random variable

We can generalize the idea of the expected value of a coin flip to all discrete random variables. The population mean of a discrete random variable is called the **expected value**.

expected value

The expected value of a (discrete) random variable X is defined to be

$$E(X) = \mu_X = \sum_{x \in \mathcal{S}_X} x \cdot P(X = x).$$

Expected Value, discrete random variable, example

What is the expected value of the random variable corresponding to rolling a die?

$$\begin{aligned} E(X) &= \sum_{x \in \mathcal{S}_X} x \cdot P(X = x) \\ &= 1 \cdot P(X = 1) + 2 \cdot P(X = 2) + \dots \end{aligned}$$

Expected Value, notes

Note about the expected value

- ▶ the most common **parameter** in statistics,
- ▶ the population analogue to sample mean; a measure of center
- ▶ this is essentially where the sample mean comes from. For a fair die or coin, we have n randomly sampled (all occur with equal probability) outcomes, so that $P(X = x_i) = 1/n$.

$$\bar{X} = \sum_{i=1}^n x_i P(X = x_i)$$

Expected Value, continuous random variable

The expected value for continuous random variables is defined similarly, but with integrals in place of sums.

$$E(X) = \mu_X = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

Variance and Standard Deviation

The population variance is defined via the expected value and the expected value property about any function h .

variance

The variance is defined to be

$$\text{Var}(X) = \sigma_X^2 = E[(X - \mu_X)^2].$$

standard deviation

The standard deviation is defined to be

$$\sigma_X = \sqrt{\sigma_X^2}.$$

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Bernoulli Distribution

The expected value of the random variable $X \sim \text{Bernoulli}(p)$ is simple to calculate. We'll look at how to calculate its expected value and variance in the next section.

x	0	1
$P(X = x)$	$1 - p$	p

Uniform Discrete

The discrete uniform distribution $X \sim \mathcal{U}(a, b)$ has PMF

$$f(x|a, b) = \frac{1}{b - a + 1} = \frac{1}{n},$$

where $n = b - a + 1$.

Uniform Continuous

The continuous uniform distribution $X \sim \mathcal{U}(a, b)$ has probability density function (PDF)

$$f(x|a, b) = \begin{cases} \frac{1}{b-a} & x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

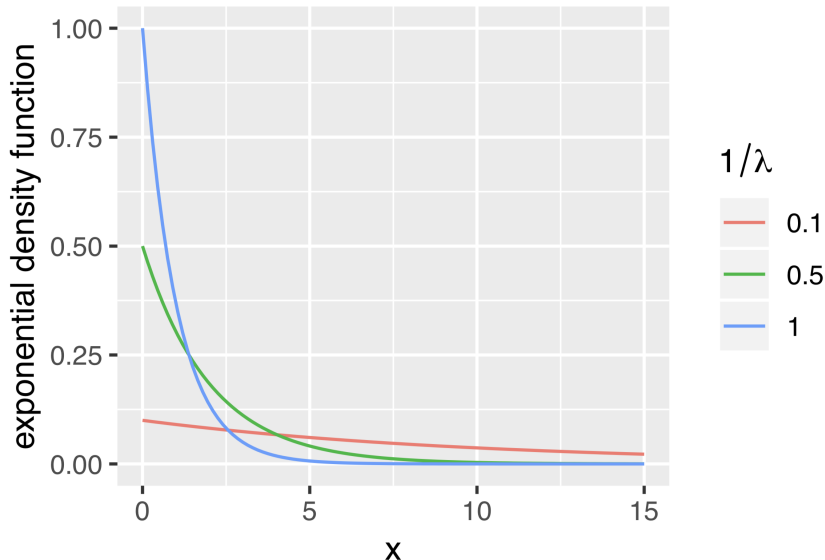
Exponential

The exponential distribution $X \sim \text{Exp}(\lambda)$ has PDF

$$f(x|\lambda) = \begin{cases} \lambda \exp(-\lambda x) & x \geq 0 \\ 0 & x < 0 \end{cases}$$

Exponential, notes

This memoryless distribution is often used to compute waiting times or survival times.



PDF, Normal

The normal (Gaussian) distribution $X \sim \mathcal{N}(\mu, \sigma^2)$ has PDF

$$f(x|\mu, \sigma) = (2\pi\sigma^2)^{-1/2} \exp \left\{ \frac{-(x - \mu)^2}{2\sigma^2} \right\}.$$

- ▶ area under the curve is equal to 1,
- ▶ unimodal (one hump)
- ▶ perfectly symmetric about μ
- ▶ parameters: centered at μ with standard deviation σ
- ▶ often $\mu = 0$, shifts the distribution
- ▶ often $\sigma = 1$, scales the distribution
- ▶ short hand $N(\mu, \sigma^2)$,
- ▶ use Z to denote standard normal distribution: $N(0, 1)$

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Parameters of Distributions

Statistics assumes the data we observe, dependent on its type and support set, follow random variables that have known functional forms with unknown parameters. From the sampled data, via the random variable's function (that we know), we estimate the parameters of the function. Specifically, we calculate the parameters that would be most likely to generate the data we saw.

Parameters of Distributions, more tangibly

If we observe data x that we assume was generated from the random variable $X \sim \text{Exp}(\lambda)$, then we can calculate the sample mean (estimating the population mean) and the sample variance (estimating the population variance).

sample (statistic)	population (parameter)
$\bar{X}$	$E(X), \mu$
$\hat{\sigma}^2$	$\text{Var}(X), \sigma^2$
$\hat{\sigma}$	σ
$\dots$	$\dots$

Expected Value, Bernoulli Distribution

The parameter of most interest with Bernoulli random variables is the probability of success p . We estimate p with $\bar{X}$, sometimes also called $\hat{p}$. Since $\bar{X}$ is the sample analogue of $E(X)$, it makes sense that $\bar{X}$ estimates p .

$$E(X) = 1 \cdot p + 0 \cdot (1 - p) = p$$

Variance, Bernoulli Distribution

The variance of the Bernoulli distribution is

$$\text{Var}(X) = p(1 - p).$$

Parameters, Exponential Distribution

- ▶ Expected value, $1/\lambda$
 - ▶ The expected value for the exponential distribution is $E(X) = 1/\lambda$. Since $\bar{X}$ estimates $E(X)$, we can assume (correctly) that $1/\bar{X}$ estimates λ .
- ▶ Variance, $1/\lambda^2$

Expected Value, Normal Distribution

- ▶ Expected value, μ
 - ▶ The expected value for the normal distribution is $E(X) = \mu$. Since $\bar{X}$ is the sample analogue of $E(X)$, we know that $\bar{X}$ estimates μ .
- ▶ Variance, σ^2

Parameters of Distributions, summary

Let's summarize the notation we just spoke about.

- ▶ Bernoulli

- ▶ $\bar{X} = \hat{p}$ estimates $E(X) = p$.
- ▶ $\hat{\sigma}^2$ (or s^2) estimates $\text{Var}(X) = \sigma^2 = p(1 - p)$.

- ▶ Exponential

- ▶ $\bar{X} = 1/\hat{\lambda}$, so $1/\bar{X}$ estimates λ .
- ▶ $\hat{\sigma}^2$ (or s^2) estimates $\text{Var}(X) = \sigma^2 = 1/\lambda^2$.

- ▶ Normal

- ▶ $\bar{X} = \hat{\mu}$, estimates $E(X) = \mu$.
- ▶ $\hat{\sigma}^2$ (or s^2) estimates $\text{Var}(X) = \sigma^2$.

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Take Away

- ▶ Random variables are the main underlying object of interest to statisticians.
- ▶ RVs allow us to calculate probability statements: from flipping a coin to probability we wait more than 15 minutes for the bus.
- ▶ RVs have distribution functions that we assume describe populations of interest.
- ▶ These distributions have known functional forms indexed by unknown parameters.
- ▶ Our goal is to estimate these parameters from sample data.

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Adapted from notes by Dr. Roualdes.